

DESIGN & FABRICATION OF LPG REFRIGERATION SYSTEM & COMPARISON WITH DOMESTIC REFRIGERATOR



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Submitted for the partial fulfillment of the Degree of Bachelor of Science in
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This report

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AUTHOR'S DECLARATION

We then declare that we are the only ones who have come up with this idea. This is a certified copy of the proposition, including any major revisions that my investigators have noticed. We gathered a lot of information from diaries and files and then developed our own strategy. We haven't taken any of the material from anywhere else, and the plan is entirely original. The supporting information is cited in the bibliography.

DECLARATION

We guarantee that this research project titled "**DESIGN & FABRICATION OF LPG REFRIGERATION SYSTEM & COMPARISON WITH DOMESTIC REFRIGERATOR**" was completed by us and is entirely our own evaluation effort. This hypothesis's work has never been published. Similarly, information derived from primary and secondary sources has been appropriately referenced and interpreted.

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LANGUAGE CORRECTNESS CERTIFICATE

The language of tables, text, and figures used throughout this thesis has been thoroughly reviewed and validated by an English linguistics expert, ensuring that it is free of syntactic, composition, grammar, and spelling problems. The criteria established by DME, FET, and IIUI guided the organisation of this thesis.

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Abstract

Power is as yet not present in a few pieces of our country. As a result, our work will be valuable for things like food refrigeration, medicines, and other similar items. This paper examines the consequence of an examination study completed to decide the presentation of the homegrown fridge. Liquified oil gas which contains propane, butane and isobutane. LPG is less expensive and have a harmless to the ecosystem nature with no ozone exhaustion and no a dangerous atmospheric deviation. LPG is utilized for cooking purposes and cooler is work on LPG. LPG is utilized as a refrigerant and cooler works effectively. LPG is more affordable and biological pleasing nature with no ozone. Fatigue potential and no an unnatural climate change potential. However, evaporator temperature accomplished from 35oc to - 5oc in 5 to 10 minutes. We may predict the best advantage of cooling sway with the appropriate operating condition of the overseeing valve and thin compartment of the structure based on the ecological examination.

We designed and tested a chiller that ran on LPG. In chambers, you can get high-compression LPG. The pressing factor of LPG is diminished due to abrupt extension and stage change of LPG occurs in an isenthalpic cycle when it passes through the hairlike container with extremely low inward breadth. The temperature declines when the fluid refrigerant transitions from a liquid to a gaseous state. LPG can provide cooling in a box in this manner.

Keywords: Vapor Absorption Refrigeration, LPG Refrigerant, Zero Refrigeration Cost, COP, Domestic Refrigerator, Eco-Friendly Refrigerants

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“All praises be to Allah, Lord of the Worlds, The Beneficent and the Merciful. Owner of the Day of Judgment, Thee (alone) we worship; Thee alone we ask for help. Show us the straight path, the path of those whom Thou hast favored; Not (the path) of those who earn Thine anger nor of those who astray”.

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Table of Contents

List of Acronyms	
xiv	
CHAPTER 1 INTRODUCTION	1
1.1 Project Title	1
1.2 Background	1
1.3 Overview:	2
1.4 Problem Identification.....	3
1.5 Project scope	4
1.6 Timeline	4
CHAPTER 2 LITERATURE SURVEY	5
2.1 Research Paper Review	5
CHAPTER 3 DESIGN OF LPG REFRIGERATION SYSTEM	8
3.1 Design of Capillary Tube	8
3.2 Design of Evaporator	8
CHAPTER 4 TYPES OF REFRIGERATION SYSTEM	9
4.1 Refrigeration & Second Law of Thermodynamics	9
4.2 Units of Refrigeration	9
4.3 Ice Refrigeration.....	10
4.4 Natural Refrigeration.	10
4.5 Cooling Via Evaporator	10
4.6 Artificial Refrigeration.....	11
4.7 Vapor Compression Refrigeration System.....	11
4.8 Domestic Refrigeration System	12
4.9 Air Conditioning System.....	14
4.10 Vapor Absorption Refrigeration System	15
4.11 Gas Refrigeration System.....	16

4.12 Steam Jet Refrigeration System	17
4.13 Dry Ice Refrigeration	19
CHAPTER 5 Working of VCR System.....	20
5.1 Comparison Between Gas & Vapor Cycle.....	20
5.2 Mechanism of Simple VCR System.....	20
5.3 Simple VCR Cycle	22
5.3.1 Compressor.....	22
5.3.2 Condenser.....	22
5.3.3 Expansion	23
5.3.4 Evaporator	23
5.4 Types of Refrigerants Used in VCR	23
5.5 Advantages and Disadvantages of VCR	24
5.6 Application of VCR	25
5.7 Temperature Zone and Rating.....	25
5.8 Fundamental of Refrigeration	25
5.9 Refrigerant.....	26
5.9.1 Primary & Secondary Refrigerants	26
5.9.2 Refrigerants Selection Criteria	27
CHAPTER 6 LPG REFRIGERATION.....	28
6.1 Physical Properties & Characteristics of LPG	29
6.2 Drawbacks of LPG	30
6.3 Components of LPG System	30
CHAPTER 7 CONSTRUCTION & WORKING.....	33
7.1 Construction of LPG Refrigeration	33
7.2 Working of LPG Refrigeration	34
7.3 Advantages & Disadvantages of LPG Refrigeration System.....	35

7.4 Application of LPG Refrigeration System	36
7.5 Result Verification	36
7.6 Conclusion.....	37
RESFRENCE	39

LIST OF TABLES

Table 1.1: Timeline.....	4
Table 5.1: Properties of Commonly Used Refrigerants.....	23
Table 5.2: Performance of Commonly Used Refrigerants.....	24
Table 7.1: Result Verification.....	37

LIST OF FIGURES

Figure 1.1: Mid 1940s Refrigerator	2
Figure 4.1: VCR System	12
Figure 4.2: Air Conditioning System.....	15
Figure 4.3: Vapour Absorption System	16
Figure 4.4: Air Cycle Refrigeration System.....	17
Figure 4.5: Steam Jet Refrigeration System	18
Figure 5.1: T-S Diagram of Simple VCR System	22
Figure 5.2: PH Diagram of Simple VCR System	22
Figure 5.3: VCR System	24
Figure 6.1: LPG Refrigeration	28
Figure 6.2: LPG Cylinder	31
Figure 6.3: Capillary Tube	31
Figure 6.4: Evaporator	32
Figure 6.5: Pressure Gauge	32
Figure 6.6: High Pressure Pipes.....	32
Figure 6.7: Regulator	32
Figure 7.1: Construction of LPG Refrigeration	33
Figure 7.2: Principle of LPG Refrigeration	35

List of Acronyms

LPG	Liquified Petroleum Gas
VCR	Vapour Compression Refrigeration
VOR	Vapour Absorption Refrigeration
HFC	Hydrofluorocarbons
CFC	Chlorofluorocarbons

CHAPTER 1

INTRODUCTION

This chapter includes

1.1 Project Title

1.2 Background

1.3 Overview

1.4 Problem Identification

1.5 Project Scope

1.6 Timeline

1.1 Project Title

Design and fabrication of LPG refrigeration system and comparison with domestic refrigerator.

1.2 Background

First cooling framework was included ice and afterward counterfeit refrigeration framework began in the 1740. In 1833 the main working VCR framework was presented. In 1854, first ice making mechanical assembly was concocted. In 1913, home use coolers was imagined. The development of freon extended the cooler market during the 1930s. Home coolers as isolated compartments were imagined in 1940. Frozen food sources, already an extravagance thing became typical.

The researcher acquainted a couple of meandering machines with accomplish some joy. In any case, it hailed the way by welcoming the consideration of researcher for legitimate examined and examination. Before the end of the nineteenth century, they had the option of building a sensible and dependable equipment for the refrigeration trades. The study of refrigeration rose to its current heights with the introduction of efficient rotary blowers and gas turbines. The Jews, Greeks, and Romans dumped a lot of snow into large pits dug into the ground and surrounded by wood and straw. The ancient Egyptians filled earthen containers with bubbling water and placed them on their rooftops, thus exposing the containers to the chilly night air. Vanishing cooling was used in the Middle East. When a fluid disintegrates quickly, it expands swiftly. The growing fume modules unexpectedly increase their active energy, which is derived from the fume's moderate ambient factors. These constituents of the environment are then cooled. Adding synthetic

compounds like sodium nitrate or potassium nitrate to water to lower the temperature was the transitory stage throughout the entire history of cooling food kinds. The above method of chilling wine was first documented around 1550. In house cooler units are utilized. Business cooler and cooler units were being used for very nearly 50 years. New coolers might comprise programmed thawing out, chilled water and ice from an allocator in the chamber.



Figure 1.1 Mid 1940s Refrigerator

1.3 Overview

The term 'refrigeration' according to an exhaustive point of view is used for the way toward taking out heat from the material. It consolidates the way toward falling and staying aware of the temperature of a body under the temperature of its ecological components. The refrigeration infers a continued with withdrawal of warmth from a body, whose temperature is as of now under the temperature of its natural components. For example, if some space (say in crisp storing) is to be kept at -2°C , we should interminably isolate warmth which streams into this is a direct result of spillage through the dividers and moreover the glow, which is conveyed into it with the articles set aside after the temperature is one lessened to -2°C . Anyway in a cooler, heat is basically being driven from a lower temperature to a higher temperature. The refrigeration structure is known to the man.

According to the second law thermodynamics, this cycle can be performed with the load of some external work. It is in this manner obvious that the load of power is regularly expected to energy a cooler. The substance which work in a glow direct to eliminate heat from an infection body and to siphon it to a hot body is characterized "refrigerant".

Cooler is disconnected in types in which each type having their own sort of working. Mechanical coolers are used as food taking care of, engineered getting ready, and cold amassing. Present day coolers, which once in a while use salt refrigeration to stay aware of temperature, is imperative for PC, food and blood antibodies that should keep a predictable and reliable temperature reliably.

It works on the standard that during the change of LPG into vaporous construction, improvement of LPG occurs. In view of this augmentation, there is a squeezing factor drop and development in volume of LPG that results in the drop of temperature and pressure and a refrigerating effect is conveyed. Refrigerating effect can be used for cooling purposes. But, this work offers refrigeration to socially pertinent necessities similarly as replaces a risky air deviation creator refrigerants. While in LPG refrigeration structure, Conventional Vapor Compression Refrigeration System uses LPG as refrigerant and formed the refrigerating effect. Notwithstanding, in our endeavor uncommonly clear sort of refrigeration structure in which high squeezing factor LPG is gone through a hairlike chamber and develops. After improvement the time of LPG changes and changed over from liquid to gas and thereafter it goes through the evaporator where it acclimatizes the glow and creates the refrigerating outcome. After evaporator it goes through the gas burner or generator where it uses further as a fuel.

1.4 Problem Identification

To enormous interest of power across the world, we consider recuperating the power which is as of now utilized yet not being used any longer, to defeat this emergency with less speculation. The climatic change and a worldwide temperature alteration request high and moderate cooling frameworks as forced air systems and fridges. Hundreds of billions of dollars are invested each year to help carry out this project. However, we recommend cooling frameworks that are free of charge. Despite the fact that government entities have not had the ability to consistently deliver a major portion of power in both metropolitan and provincial regions. Individuals in such areas require refrigeration for a variety of socially significant functions like as food storage, storing clinical supplies, and homegrown kitchens. This project has the capability of using LPG instead of electricity for refrigeration. This setup works well for refrigeration in locations where there is a power shortage.

1.5 Project Scope

- This project has a main scope in restaurants of Europe and vehicles.
- Our LPG refrigeration system is very low cost, durable and also easy to install. It would also be helpful for public/private sector.
- We can also further recommend this project for business purpose

1.6 Timeline

Table 1.1 Timeline

<i>Sr. No.</i>	<i>Task</i>	<i>Time</i>
<i>1</i>	<i>Project Identification</i>	<i>September</i>
<i>2</i>	<i>Literature Review</i>	<i>October</i>
<i>3</i>	<i>Market Survey</i>	<i>November</i>
<i>4</i>	<i>Fabrication</i>	<i>December + January</i>
<i>5</i>	<i>Assemble</i>	<i>February</i>
<i>6</i>	<i>Final Experiment</i>	<i>March + April</i>
<i>7</i>	<i>Thesis Completion</i>	<i>June</i>

CHAPTER 2

LITERATURE SURVEY

This chapter includes

2.1 Research Paper Review

2.1.1 Smyth et.al.

Direct expansion (DX) refrigeration technique is almost entirely employed in multitemperature transport refrigeration systems, according to Smyth et al. To function across a wide variety of set-point conditions, multitemperature systems require up to three evaporators, necessitating substantial refrigerant charges and system pressure control. Despite incremental design advances over the last decade, DX systems continue to have environmental and control difficulties. The use of indirect refrigeration systems (IDX) as a solution to these problems is a viable option. Indirect systems, on the other hand, may experience performance penalties due to reduced cooling capacity and coefficient of performance (COP) under specific operating situations. Incorporating an economizer circuit into the IDX system's primary cycle is one technique for mitigating the disadvantage of limited refrigeration capacity. Economizer cycles can improve the refrigeration effect of the primary refrigerant in the indirect system's primary to secondary heat exchanger.

2.1.2 A. Baskaran & P. Koshy Mathews

A Performance Comparison of Eco-Friendly Vapor Compression Refrigeration Systems VCR system with the new R290/R600a refrigerant with minimal global warming potential as a CFC12 and HFC 134a replacement refrigerant The R290/R600a refrigerant has a refrigerating capability of 28.6 percent to 87.2 percent higher than R134a.

2.1.3 M. Mohanraj Et. Al.

In a household refrigerator, a drop-in replacement for R134a was tested with an ecologically friendly, energy efficient hydrocarbon (HC) mixture of 45 percent HC290 and 55 percent R600a at various mass charges of 50g, 70g, and 90g. The experiments were conducted out in a 165-liter family refrigerator using R134a as the lubricant and POE oil as the lubricant. The discharge temperatures of HC mixes are found to be 13.76%, 6.42%, and 3.66 percent lower than R134a for 50g, 70g, and 90g, respectively. At 50g and 70g, the power consumption of the HC combination is 10.2 percent lower and 5.1 percent, respectively,

with 90g consuming 1.01 percent more electricity. When compared to R134a, the percentage reduction in pull down time is 18.36 percent, 21.76 percent, and 28.57 percent for 50, 70, and 90g mass charges, respectively. Because of its excellent energy efficiency, the HC combination will help lessen indirect global warming. Finally, in a 165-liter household refrigerator, a 70-gram HC combination was proven to be a viable alternative to R134a.

2.1.4 B. O. Bolaji

In a family refrigerator, researchers tested R152a/R32 as a replacement for R134a and determined that R152a has a COP of 4.7 percent higher than R134a. R32 has a lower coefficient of performance (COP) than R134a, and propane is a more appealing and environmentally friendly alternative to the CFCs now in use.

2.1.5 R. W. JAMES & J. F. Missenden

Analysis the implications of using propane in household refrigerators in terms of consumption, compressor prices, availability, energy lubrication, environmental issues, and safety. Propane is a viable and environmentally friendly replacement for the CFCs now in use.

2.1.6 Bilal Et. Al.

He has undertaken experiments on the performance of liquefied petroleum gas (LPG) as a prospective R12 replacement in household freezers. The experiment for LPG (30 percent propane, 55 percent n-butane) is being conducted in a refrigerator that was originally meant to function with R12. And Isobutane (15% isobutane). During the test, mass loads of 50, 80, and 100 grammes of LPG were used. R12 and LPG are quite similar. At evaporator temperatures below 15°C, the COP was higher for all mass charges. Overall, it was discovered that while using this refrigerator with an 80g charge, LPG produced the best results. The temperature of the condenser was kept constant at 47 degrees Celsius. Cooling capabilities were measured, and they were found to be three to four times higher for LPG than for R12.

2.1.7 M. Fatouh Et. Al.

Utilized Liquefied petrol gas (LPG) comprised of 60% propane and 40% business butane as a substitute for R134a in a solitary evaporator private cooler with an all out volume of 0.283

m3. Distinctive slim lengths and temperatures were utilized to test the fridge's presentation. R134a and LPG have distinctive valuing. The fridge's trial discoveries utilizing 60 g of LPG and a narrow cylinder length of 5 m were contrasted with those utilizing 100 g of R134a and a slim cylinder length of 4 m. LPG coolers had more limited draw down occasions, lower pressure proportions, and lower power use. 7.6 percent, 5.5 percent, and 4.3 percent, individually, of R134a. The COP of a LPG cooler was 7.6% more prominent than that of an electric fridge. R134a is a sort of refrigerant. Contrasted with R134a, the LPG cooler had a lower on-time proportion and energy utilization of 14.3 percent and 10.8 percent, separately. To outlines, the proposed LPG is a drop-in substitution for R134a; by and by, to accomplish further developed execution, narrow length and refrigerant charge improvement were required.

CHAPTER 3

DESIGN OF LPG REFRIGERATION SYSTEM

This chapter includes

3.1 Design of Capillary Tube

3.2 Design of Evaporator

3.1 Design of Capillary Tube

In refrigeration and air conditioning systems, the capillary tube is one of the most common throttling devices. Capillary tubes are made of copper tubing with a very small internal diameter. It's long and coiled into several turns to save space. Internal diameters of capillary tubes used in refrigeration and air conditioning applications range from 0.5 to 2.28 mm (0.020 to 0.09 inches). A capillary tube is used as a throttling device in home refrigerators, deep freezers, water coolers, and air conditioners.

The following two elements contribute to the pressure drop through the capillary tube:

1. Friction causes frictional pressure drop due to fluid viscosity.
2. The flashing of the liquid refrigerant into vapor causes acceleration and a drop in momentum pressure.

The following are the design parameters for a capillary tube:

Cylinder Size = 5 kg,

Diameter of Cylinder = 190mm

Diameter of Capillary = 1mm

3.2 Design of Evaporator

The evaporator is a refrigeration system component that removes heat from air, water, or any other body that needs to be cooled by the evaporating refrigerant. Plate and tube type evaporators were utilized in the experiment because they give a gentle sort of evaporation with a short residence period. It also protects food and other products from bacterial attack and is inexpensive to install. The primary function of evaporator coils is to transform the refrigerant from a liquid to a gaseous condition. The refrigerant must travel from the expansion valve to the evaporator coils in order to complete the operation. The refrigerant pressure drops as a result of the movement, and the expansion valve converts the liquid condition to a gaseous

CHAPTER 4

TYPES OF REFRIGERATION SYSTEM

This chapter includes

- 4.1 Refrigeration & Second Law of Thermodynamics
- 4.2 Units of Refrigeration
- 4.3 Ice Refrigeration
- 4.4 Natural Refrigeration
- 4.5 Cooling Via Evaporator
- 4.6 Artificial Refrigeration
- 4.7 Vapor Compression Refrigeration System
- 4.8 Domestic Refrigeration System
- 4.9 Air Conditioning System
- 4.10 Vapor Absorption Refrigeration System
- 4.11 Gas Refrigeration System
- 4.12 Steam Jet Refrigeration System
- 4.13 Dry Ice Refrigeration

4.1 Refrigeration and Second Law of Thermodynamics

Refrigeration is the way toward bringing down and keeping a temperature beneath that of the encompassing air, determined to cool an item or spot to the ideal temperature. The conservation of transient food merchandise by putting away them at low temperatures has been quite possibly the main utilizations of refrigeration. Refrigeration frameworks are likewise generally utilized to give warm solace to individuals through cooling. Cooling is the way toward getting air control temperature, dampness content, tidiness, scent, and dissemination simultaneously, as needed by occupants, cycles, or items in the space. Refrigeration and cooling have a long history, coming from human requirements for food and solace. The historical backdrop of refrigeration is interesting since it covers each feature of the business, including refrigerant accessibility, main players, blower headways, and refrigeration frameworks.

4.2 Units of Refrigeration

The Ton of Refrigeration is a unit used in the refrigeration industry. A tonne of refrigeration is defined as the amount of heat that must be removed from one tonne of ice in 24 hours when the underlying temperature of the water is 0 °C, on the assumption that dissolving a same amount of ice will provide an identical cooling effect.

4.3 Ice Refrigeration

Ton of Refrigeration is a unit of measurement used in the refrigeration industry. When the underlying state of water is 0 °C, a tonne of refrigeration is defined as the amount of heat that must be removed from one tonne of ice within 24 hours, on the basis that dissolving a similar ice will provide an identical cooling effect.

4.4 Natural Refrigeration

Natural methods of refrigeration, such as the usage of ice or evaporative cooling, were used in the past.

Ice used to be one of two things:

Transported from colder places,

Produced at night by cooling water through radiation to the stratosphere, or harvested in the winter and stored in ice houses for use in the summer.

A variety of icehouses were erected in Europe, America, and Iran to store ice. In these icehouses, insulating materials such as sawdust or wood shavings were used. Cork was later employed as an insulating substance. Ice has always been available to the aristocratic who could afford it according to literature. In America, the ice trade was a thriving industry. Ice was delivered to southern states in the United States in train compartments insulated by 0.3m of cork insulation. In addition to the United States, ice trade was popular in a number of other nations. In these countries, ice was either imported from colder climates or collected in the winter and stored in icehouses for use in the summer.

The ice trade when the United States alone exported 225,000 tons of ice to many countries.

However, when artificial refrigeration became more common, the ice trade began to dwindle.

4.5 Cooling Via Evaporator

The technique of reducing a system's temperature by evaporating water is known as evaporative cooling. If the temperature of the environment is higher than the temperature of your skin, humans perspire and use evaporative cooling to dissipate metabolic heat. For evaporative cooling, animals like the hippopotamus and buffalo wrap themselves in mud.

Evaporative cooling is a type of environment cooling technique that uses water evaporation to cool the surrounding air. It is utilized to provide air cooling and conditioning services in data centers and major IT facilities. Evaporative cooling is one of the most simple and environmentally friendly ways to keep data centres cool.

Evaporative cooling happens when water is pushed onto a cooling pad until it is completely saturated.

A fan placed near the cooling pad draws air in and blows it out into the surrounding environment. In delivering air conditioning services, the evaporative cooling technology uses less energy than conventional data center cooling solutions.

Swamp cooling, desert cooling, and wet air cooling are all terms for evaporative cooling.

4.6 Artificial Refrigeration

Refrigeration as we realize that it is accomplished by counterfeit strategies. However recognizing normal and counterfeit refrigeration is troublesome, it is by and large concurred that fake refrigeration started in 1755, when Scottish scholarly William Cullen made the exceptionally refrigerating framework fit for delivering a humble measure of ice in the lab. Refrigeration frameworks are classed as fume pressure frameworks, fume ingestion frameworks, gas cycle frameworks, etc, contingent upon their working standard.

4.7 Vapour Compression Refrigeration

The capacity of fluids to ingest colossal measures of warmth when they bubble and vanish is the establishment of contemporary refrigeration. In 1755, Professor William Cullen showed this by placing some water in warm contact with ether under a vacuum siphon's recipient. As a result of the vacuum siphon, the pace of ether dissipation expanded, and water may freeze. The fume pressure and inactive warmth are two thermodynamic ideas associated with this interaction. At a pressing factor called the immersion pressure, a fluid is in warm balance with its own fume at a pressing factor that is exclusively controlled by temperature. At the point when the pressing factor in a pressing factor cooker is expanded, the water bubbles at a more noteworthy temperature. The subsequent idea is that fluid vanishing requires the arrival of inert warmth. At the point when the fluid's dormant warmth is eliminated, the fluid cools. However long the vacuum siphon keeps a pressing factor equivalent to immersion pressure at the predefined temperature, the temperature of ether will stay steady.

This necessitates the elimination of all fumes produced by vaporization. If a lower temperature is wanted, the vacuum pump will have to maintain a lower saturation pressure.

The evaporator is a component of a modern refrigeration system that uses this technology to produce cooling. If this cooling process is to be maintained, the vapours must be recycled to the liquid form through condensation.

Condensation necessitates heat rejection to the environment. At room temperature, it can be condensed by increasing the pressure.

In the second half of the eighteenth century, condensation was discovered. U.F. was used to liquefy SO_2 . G. Clouet and Clouet are two authors who have collaborated on a Van Marum and Van Frostwork liquefied NH_3 in 1787, although Monge liquefied NH_3 in 1780. As a result, a compressor is required to maintain a high enough pressure for the evaporating vapours to condense at a temperature above that of the surrounding air.

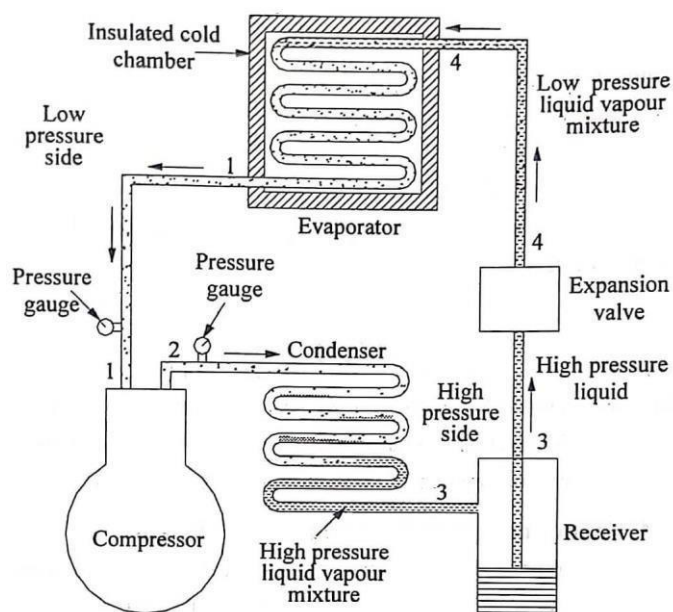


Figure 4.1 VCR System

4.8 Domestic Refrigeration System

The family cooler with regular ice (homegrown refrigerator) was created in 1803 and has stayed unaltered. Homegrown coolers used to be made of wood and protected appropriately. Ice used to be kept at the highest point of the container, bringing about low temperatures in the crate because of regular convection heat move from the ice. The water made by the dissolving of ice is gathered in a trickle dish. When the entirety of the ice has dissolved, the case should be supplanted with new ice. In spite of its essential plan, the homegrown cooler experienced various disadvantages. The ice should be recharged as it is utilized, hence the least temperatures that might be created inside the compartment are restricted. Besides, apparently warm winters in the United States have brought about a significant lack of regular ice. Thus, since 1887, endeavors have been made to build private

coolers that utilization mechanical components. Homegrown mechanical fridges were first costly, not completely programmed, and questionable. The advancement of little blowers, programmed refrigerant controls, better shaft seals, enhancements in electrical force frameworks, and acceptance engines, nonetheless, empowered the largescale improvement of mechanical private coolers. The main family cooler was presented by General Electric in 1911, trailed by Frigidaire in 1915. In 1918, Kelvinator presented the primary family mechanical fridge in the United States.

4.9 Air Conditioning System

The basic parts of a vapour pressure refrigeration framework are displayed in Figure 4.8. The essential framework incorporates of an evaporator, blower, condenser, and extension valve, as demonstrated in the graph. The refrigeration impact is accomplished in the virus zone by extricating heat from the refrigerant vaporization in the evaporator. The blower packs the refrigerant fume from the evaporator to a high pressing factor at which its immersion temperature is higher than the encompassing or some other warmth sink. Subsequently, when high-pressure, hightemperature refrigerant goes through the condenser, heat dismissal to the warmth sink causes buildup of the fume into fluid. The high-pressure fluid is compelled to course through a development valve to finish the cycle. The pressing factor and temperature of the refrigerant fall as it goes through the development valve. In the evaporator, this low-pressure, low-temperature refrigerant fume vanishes, engrossing warmth from the virus zone. It's significant that the framework runs in a shut circle. Mechanical work is needed for the framework to work. It ingests heat from a crisp climate and sends it to a high-temperature heat sink.

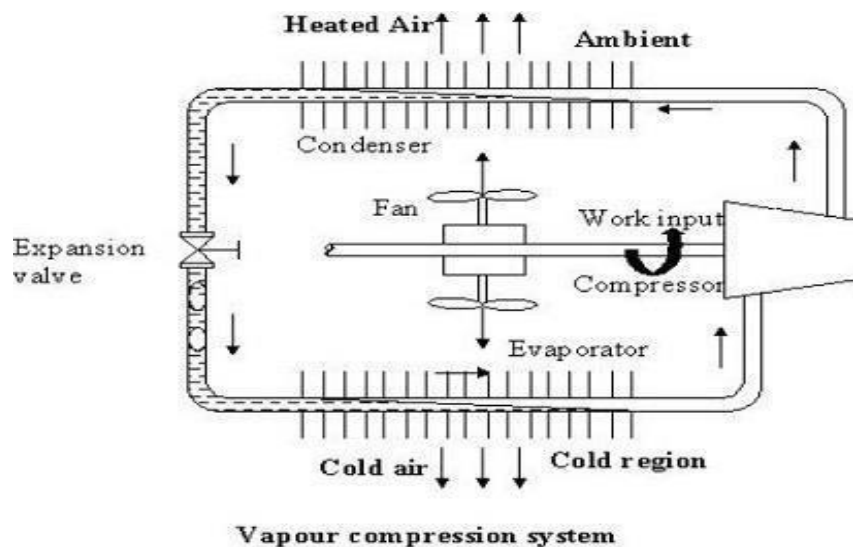


Figure 4.2 Air Conditioning

4.10 Vapour Absorption Refrigeration System

In 1810, John Leslie kept H_2SO_4 and water in two separate containers that were associated. H_2SO_4 has an exceptionally high Sulfur content. Water is one of my number one things. It catches water fume and uses this as the reason for removing the dissipated water. There is no requirement for a blower or a siphon when utilizing water fume. In this framework, H_2SO_4 is a spongy that should be utilized for nonstop working, the retained water fume is recovered by warming. This standard was utilized by Windhausen in 1878 for a retention refrigeration framework that ran on H_2SO_4 . In 1860, Ferdinand Carre concocted the water smelling salts retention strategy. Water is an astounding NH_3 safeguard. In case NH_3 is put away in a vessel that is presented to another vessel containing water, the high ingestion capability of water will incite NH_3 to vanish, dispensing with the requirement for a blower to drive the fumes. To improve the pressing factor of a solid arrangement, a fluid siphon is used. To isolate the water from the smelling salts, the solid arrangement is warmed in a generator and afterward went through a correction section. Following that, the smelling salts fume is consolidated and reused. Subsequently, the framework capacities for the most part on poor quality energy, which is used to warm the solid arrangement that isolates the water from the alkali. Initially, these frameworks were fueled by steam. Afterward, frameworks dependent on oil and gaseous petrol were taken on. The basic segments of a fume ingestion refrigeration framework are displayed in Figure 4.9. Balzar von Platen and Carl Minters,

two understudies at Stockholm's Royal Institute of Technology, fostered a three-liquid framework that didn't need a siphon in 1922. Solid and feeble arrangements were flowed utilizing a warming based air pocket siphon, and hydrogen was utilized as a non-condensable gas to bring down the halfway pressing factor of NH_3 in the evaporator.

Goeppert proposed this unique thought in 1899, anyway he was fruitless since he utilized air as a non-condensable gas. In some specialty applications, for example, lodgings, Platen-Minters refrigeration frameworks are still regularly used.

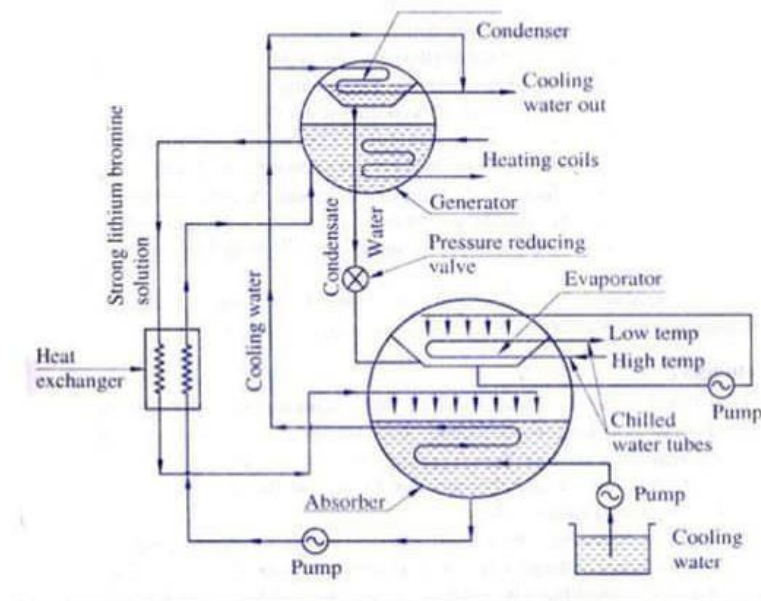


Fig 4.3 VAR System

Another possibility is a Lithium Bromide water-based vapour absorption device. This is how the chilled water air-conditioning system works. This machine is a descendent of Windhausen's, only it uses LiBr instead of H₂SO₄. LiBr is the absorbent in this system, whereas water is the refrigerant. This system makes advantage of vacuum pressures. The condenser and generator are housed in one cylindrical vessel, while the evaporator and absorber are housed in another cylindrical vessel.

This uses low-grade energy as well and requires the use of a boiler or process steam.

4.11 Gas Refrigeration System

At the point when high-pressure air extends and tackles job (like moving a cylinder or pivoting a turbine), its temperature drops. This data has been known to man since the eighteenth century. In 1807, Dalton and Gay Lusaac researched this. In 1824, Sadi Carnot depicted this as a notable wonder. Notwithstanding, in 1844, Dr. John Gorrie, a Florida doctor, created one of these machines to make ice for his fever-stricken patients. This machine made salt water at a temperature of -7°C utilizing packed air at 2 atm pressure, which was then used to make ice. An air cycle cooling system was developed by Alexander Carnegie Kirk in 1862.

This present framework's blower was controlled by a steam motor. Utilizing a Kirk could create temperatures as low as 40oC with a pressure proportion of 6 to

8. The open type of machine was consummated by Paul Gifford in 1875. T B Lightfoot, A Haslam, and others improved this machine. James Coleman and Henry Bell. For quite a while, this was the essential method of cooling ships adrift. In New York, Frank Allen created a shut cycle machine that utilizes high pressing factors to abbreviate the measure of time it takes to finish a cycle stream rates in volume. Thick air machine was the name given to it. Air cycle refrigeration is presently only utilized in airplane with super blowers fit for taking care of high volume stream rates. The schematic of an open sort air cycle refrigeration framework is displayed in Figure 4.10. A blower, an expander, and a warmth exchanger make up the essential framework envisioned here. The blower packs the air from the cold chamber. The warmed, high-pressure air in the warmth exchanger rejects warmth to the warmth sink (cooling water). In the expander, warm however high-pressure air extends. After development, the virus air is conveyed to the virus space to give cooling. Since crafted by development to some extent makes up for the work of pressure, the expander and blower share an equivalent shaft.

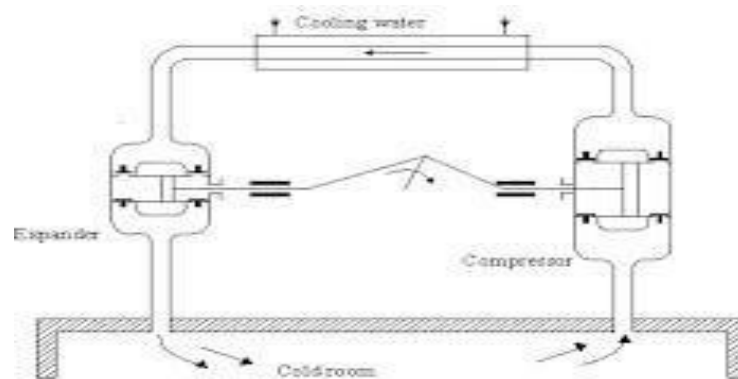


Figure.4.4. Schematic of a basic open type air cycle refrigeration system

4.12 Steam Jet Refrigeration System

Exactly when water is sprinkled into a chamber with a low squeezing factor, a piece of the water disseminates. At the squeezing factor in the chamber, the enthalpy of dissemination

will cool the abundance water to its submersion temperature. Typically, a lower temperature requires a lower pressure. Since water freezes at 0°C , temperatures under 4°C can't be cultivated using it. High velocity steam is used to entrain the scattering water smolder in this structure. High-pressure goal steam adventures through a centered or joined divergent spout, achieving sonic or supersonic speed and a low strain on the solicitation for 0.009 kPa , contrasting with a 4°C evaporator temperature. The massive energy of reasoning steam entrains or takes the evaporating water smolder from the gleam chamber with it. It moves the exhaust against the squeezing factor tendency up to the condenser, where the squeezing factor is $5.6\text{-}7.4\text{ kPa}$ and the temperature is $35\text{-}45^{\circ}\text{C}$. Both the aim and utilized tilize are thick and recuperated in this cycle. Steam fly refrigeration is the name for this methodology.

A schematic of the framework is displayed in Figure 4.11. As can be seen, this framework requires the upkeep of a decent vacuum. Supporter ejectors are in some cases utilized for this reason.

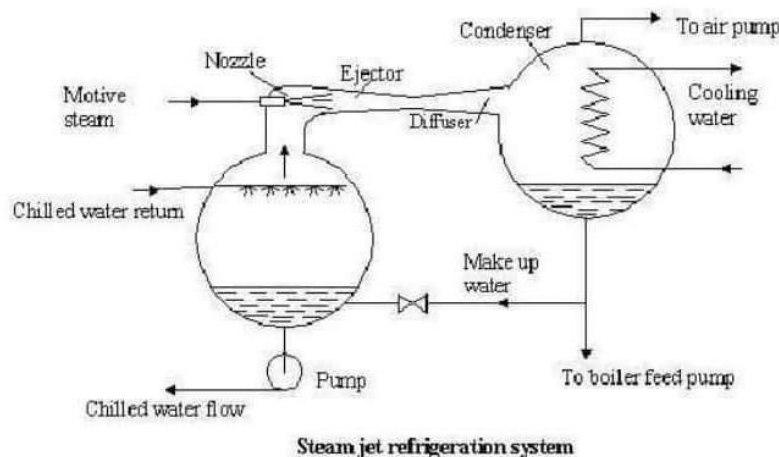


Figure.4.5. Schematic of a Steam Jet Refrigeration System

Pelletan, a Frenchman, was granted a patent in 1838 for compressing steam using a jet of motor steam. Charles Parsons, an Englishman, investigated the potential of lowering pressure via the entrainment effect of a steam jet around 1900. The steam jet refrigeration technology, on the other hand, is credited to Maurice Leblanc, a French engineer who developed it in 1907-08. Ejectors were employed to create a high-velocity steam jet (about 1200 m/s) in this system. Based on Leblanc's concept, Westinghouse developed the first commercial system in Paris in 1909. Despite the low efficiency of the steam jet refrigeration system, it was appealing since water is non-toxic and the system can run on exhaust steam from a steam engine. Breweries, chemical industries, and other businesses used stem jet refrigeration

systems. And battleships from 1910 forward. Follain, a French engineer, enhanced the system in 1926 by integrating numerous phases of suction steam vaporization and condensation. This type of system sparked a lot of interest in the United States between 1928 and 1930. In the United States, they were mostly utilized to cool factories, movie theatres, ships, and even railroad wagons.

From 1930, a number of businesses, began commercial manufacturing of these systems. However, more effective vapour absorption systems based on LiBr-water gradually superseded these systems. Even as late as the 1960s, several countries were producing these systems. Stem jet refrigeration systems were utilized in breweries, chemical plants, and other companies. I.S. Badylkes, a Russian engineer, is credited with designing closed vapour jet refrigeration systems for the first time around 1955. With a single stage of compression, temperatures as low as -100°C can be achieved using refrigerants other than water. The advantages claimed for this type of system include simplicity and robustness, whereas the main downsides are complicated design and cost.

4.13 Dry Ice Refrigeration

Dry ice is a highly concentrated carbon dioxide. The property of dry ice to shift directly from solid to fume state is used to keep up with low temperatures. Dry ice is commonly used to fill frozen food containers. As it decomposes into a fume, dry ice keeps the food frozen.

CHAPTER 5

WORKING OF VCR SYSTEM

This chapter includes

5.1 Comparison Between Gas & Vapor Cycle

5.2 Mechanism of Simple VCR System

5.3 Simple VCR System

5.3.1 Compression Process

5.3.2 Condensing Process

5.3.3 Expansion Process

5.3.4 Vaporizing Process

5.4 Types of Refrigerants Used in VCR

5.5 Advantages and Disadvantages of VCR

5.6 Application of VCR

5.7 Temperature zone and rating

5.8 Fundamental of Refrigeration

5.9 Refrigerant

5.9.1 Primary & Secondary Refrigerants

5.9.2 Refrigeration Selection Criteria

5.1 Comparison Between Gas Cycles and Vapor Cycles

The two types of thermodynamic cycles are gas and vapour cycles. The working fluid (a gas) does not undergo phase shift in a conventional gas cycle, hence the operational cycle will be disrupted by the vapour dome. A gas cycle occurs when a gas is subjected to heat rejection and cooling. Thermodynamically cooling and heating the working fluid undergoes phase change in a vapour cycle. The evaporation of the refrigerant liquid produces the cooling effect. The temperature of the refrigerant remains constant during phase transition operations if it is a pure material. As a result, the required mass flow for a given refrigeration capacity will be significantly lower when compared to gas cycle rates. Vapour compression systems, vapour absorption systems, for example, can all be subdivided into vaporisation cycles. The most common type of refrigeration is vapour compression.

5.2 Mechanism of Simple Vapor Compression Refrigeration System

Pressure Refrigeration Cycles exploit the way that when emphatically packed liquids at a particular temperature are permitted to extend, they will in general become cooler. If the pressing factor change is sufficiently large, the compressed gas will be more smoking than

our source of cooling (for example, outside air) and the stretched gas will be cooler than our target chilly temperature. Liquid is used to cool a low-temperature climate while dismissing warmth to a hightemperature climate in the present circumstance. There are two benefits to utilizing a fume pressure refrigeration cycle. First off, changing a fluid to a fume requires a great deal of nuclear power, hence a ton of warmth might be cleared from the cooled room. Second, in a Simple Vapor Compression Refrigeration System, the isothermal quality of vaporisation allows warmth to be extracted without raising the temperature of the functioning liquid to the temperature of whatever is being cooled. This implies that warmth move rate stays high on the grounds that the nearer the functioning liquid temperature moves toward that of encompassing the lower the pace of warmth move.

Different stages of Refrigeration cycle are:

- **1-2 Low Pressure Liquid Refrigerant**

The evaporator absorbs the heat from its surroundings, which are typically air, water, or another interacting fluid. It converts from a fluid to a gas during this contact, and the temperature at the evaporator output is somewhat superheated.

- **2-3 The Superheated Vapour**

Enters the blower, which raises the pressing factor. Since a part of the energy utilized in the pressure cycle is passed to the refrigerant, the temperature will rise.

- **3-4 High Pressure Superheated Gas**

The air movement from the blower to the condenser. The cooling cycle's first step (3-3a) superheats the gas before converting it to a fluid (3a-3b). This cycle is cooled as much as possible after it is completed, using air or water as a source of energy. The refrigerant fluid is sub-cooled as it enters the extension gadget because the temperature lowers much greater in the line work and fluid beneficiary (3b - 4).

- **4-1 High Pressure Sub Cooled Liquid**

Goes through the development gadget, which lessens the pressing factor and manages the stream into the evaporator.

5.3 Simple Vapor Compression Refrigeration Cycle

It is shown on T-S below at point 1; let T_1 , P_1 , and s_1 be the properties of the fume refrigerant; the four cycles are as follows:

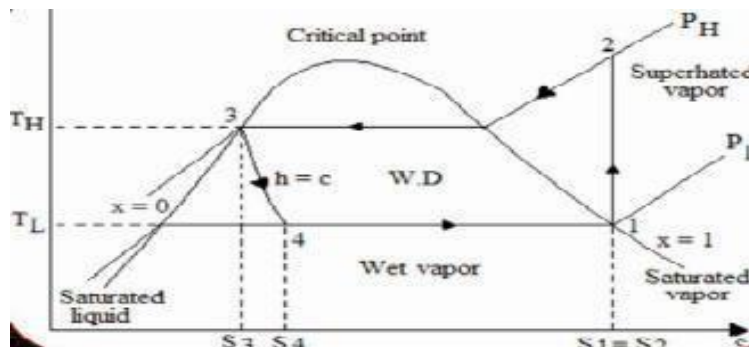


Figure 5.1 T-S Diagram of Simple VCR System

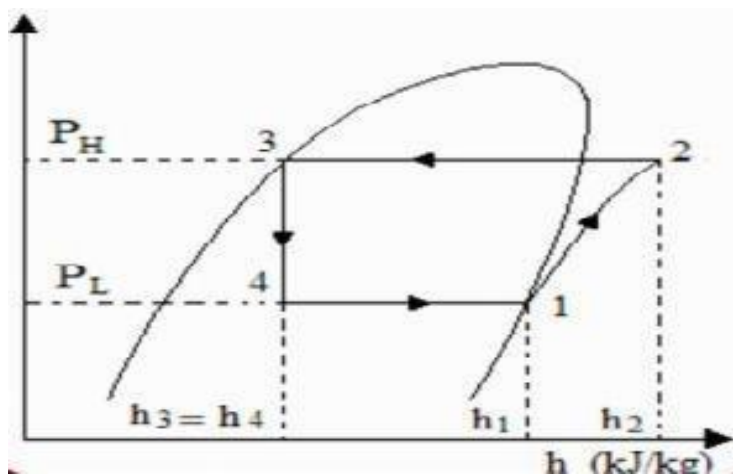


Figure 5.2 PH Diagram of Simple VCR System

5.3.1 Compression Process

The fume refrigerant is packed isentropically to dry soaked fume at low pressing factor p_1 and temperature T_1 , as proven by the upward line 1-2 on the T-s chart and the bend 12 on the p-h outline. The temperature and pressing factor ascend from 1 to 2.

$$W = h_2 - h_1$$

is the work done during isentropic compression.

5.3.2 Condensing Process

The condenser collects the high-pressure, high-temperature fume refrigerant from the blower.

In a condenser, it is entirely consolidated at a constant pressing factor p_2 and temperature T_2 . The fluid refrigerant replaces the fume refrigerant. The refrigerant produces as it passes through the condenser. Its inert warmth is transferred to the accumulating medium that surrounds it.

5.3.3 Expansion Process

Through the extension valve, the fluid refrigerant was constricted to a low pressing factor $p_4=p_1$ and temperature $T_4=T_1$ at pressure $p_3=p_2$. While some of the fluid refrigerant is dissipated as it passes through the extension valve, the majority of it is disintegrated by the evaporator. Throughout the choke contact, the fluid refrigerant retains or rejects no warmth.

5.3.4 Vaporizing Process

The refrigerant's fluid fume blend is dissipated at pressure $p_4=p_1$ and temperature $T_4=T_1$. At consistent pressing factor and temperature, the fluid refrigerant is changed over to a fume refrigerant. The fluid dissipates all through the cycle. The inactive warmth of vaporization is consumed by the fume refrigerant from the medium (air, water, or brine) to be cooled. During dissipation, the warmth assimilated or separated by the fluid fume refrigerant is given by:

$$RE = h_1 - h_4 = h_1 - hf_3$$

At T_3 , hf_3 is detectable warmth (enthalpy of fluid refrigerant leaving the condenser). The worth of the coefficient of execution is the proportion of refrigerating impact to the work done.

$$\begin{aligned} C.O.P &= (h_1 - h_4) / (h_1 - hf_3) \\ &= (h_1 - hf_3) / (h_2 - h_1) \end{aligned}$$

5.4 Types of Refrigerants Used in Vapor Compression System

Vapor compression systems use a number of refrigerants. The choice of fluid is mostly determined by the needed cooling temperature. The following chlorinated fluorocarbons (CFCs, sometimes known as Freons) are commonly used refrigerants: R-11, R-12, R-21, R-22, and R-5.

Table 5.1 summarizes the properties of these refrigerants and Table 5.2 summarizes the performance of these refrigerants.

Table 5.1 Properties of commonly used refrigerants

Refrigerant	Boiling Point (C)	Freezing Point (C)	Vapor Pressure (kpa)	Vapor Volume (m ³ /kg)	Enthalpy	
					Liquid (kj/kg)	Vapor (kj/kg)
R-11	-22.79	-110.1	24.76	0.52280	189.45	390.65
R-12	-27.80	-159.1	221.30	0.066	190.56	355.43
R-22	-39.65	-159.5	355.63	0.055	185.4	411.65

R-502	-43.38	-----	421.22	0.0321	177.6	352.76
R-7 ammonia	-32.29	-76.9	277.42	0.3285	707.6	455.76

Table 5.2 Performance of commonly used refrigerants

Refrigerant	Evaporating pressure (kpa)	Condensing Pressure (kpa)	Pressure Ratio	Vapor Enthalpy (kJ/kg)	COP carnot
R-11	19.4	114.7	7.59	164.7	6.76
R-12	183.7	645.6	3.09	144.8	3.68
R-22	286.5	1324.7	3.98	155.3	3.76
R-502	350.7	1333.8	2.95	114.6	3.64
R-717	235.8	1256.4	5.67	113.2	5.74

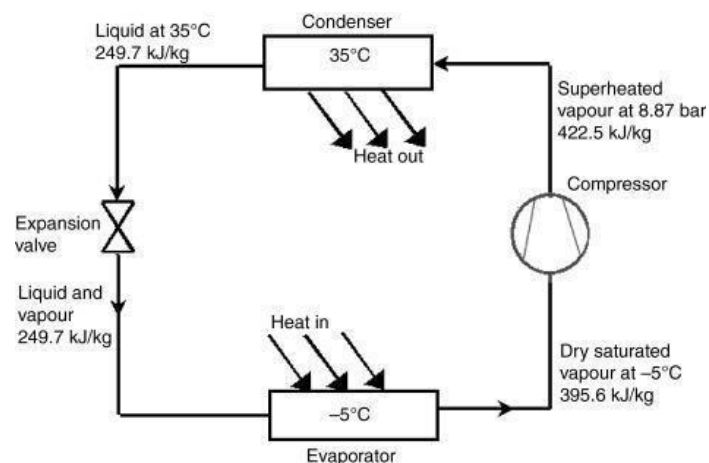


Figure 5.3 VCR System

5.5 Advantages and Disadvantages

Advantages of VCR

- High refrigerant capacity for a minimal initial investment and operating cost.
- Highly efficient.
- For little to major disaster, a very simple framework is used.
- For heat syphon activity, the cycle can be reversed.

Disadvantages of VCR

- There are clamours as a result of the system.
- Breakdowns in the refrigerant system are a possibility.
- Operators working in a limited area

5.6 Application of VCR

- Air conditioning
- House hold refrigerator
- Deep freezers

5.7 Temperature zone and Rating

To keep different types of food, some refrigerators are now separated into four zones:

- -18 degrees Celsius (0 degrees Fahrenheit) (freezer)
- 0 degrees Celsius (32 degrees Fahrenheit) (meat zone)
- 5 degrees Celsius (41 degrees Fahrenheit) (cooling zone)
- 10 degrees Celsius (50 degrees Fahrenheit) (crisper)

The capacity of a refrigerator is measured in litres or cubic feet. A linked fridge cooler's volume is often split between the cooler and the cooler, with the cooler being 100 litres (3.53 cubic feet) and the cooler being 140 litres (4.94 cubic feet).

5.8 Fundamental of Refrigeration

Refrigeration

The technique of eliminating heat from a room at a lower temperature than the ambient temperature is known as refrigeration. The natural periodicity of heat flowing from higher to lower temperatures causes the rushes to replace the heat loss. The heat that enters the system is returned to the atmosphere in a refrigerator. As a result, according to the second law of thermodynamics, systems that pump heat from lower to higher temperatures require work input.

Second Law of Thermodynamics

A cutoff law is the second law of thermodynamics. It decides a framework's greatest proficiency. The subsequent law in like manner perceives that cycles follow a particular way however not the other way. It's also how the considerable entropy feature is described. Warmth does not transfer from a body at a lower temperature to a body at a higher temperature in the same way that ordinary information does. To constantly transport heat from a lower temperature to a higher temperature, a refrigeration framework is required, which requires effort from an external source (that is, to keep up with the low temperature). This is the Clausius adaption of the subsequent law, which is one of thermodynamics' subsequent law's standards.

Clausius' Statement of Second Law

In a cyclic process, it is impossible to transport heat from a low temperature to a high temperature without the help of an external source of energy. Heat transfer can potentially spread all of the energy provided to a system as work. The energy transported as heat, on the other hand, cannot be converted into work indefinitely, resulting in a thermal efficiency of 100%. Only a fraction of heat transfer at high temperatures can be converted into work in a cyclic process; the rest must be rejected to the environment at lower temperatures. An automobile could run on free energy from the atmosphere if continuous work could be obtained by heat transfer from a single heat source. Perpetual is the name of a potential second-generation motion machine that could achieve it. This information can be found in the KelvinPlanck Statement of the Second Law.

5.9 Refrigerant

The refrigeration system's thermodynamic efficiency is primarily determined by its operating temperatures. However, fundamental practical considerations such as system design must be addressed. Size, initial and ongoing expenditures, safety, and design service ability and other factors are highly dependent on the type of vehicle. The refrigerant chosen for a certain application owing to a number of environmental concerns, such as the ozone layer. The refrigerant selection has recently become one of the most serious topics because it may necessitate many changes in refrigeration system, design and production. As a result, it's critical to comprehend the challenges surrounding refrigerant selection and application. Any fluid can theoretically be used as a refrigerant in vapour compression refrigeration systems.

5.9.1 Primary and Secondary Refrigerants

Essential and optional refrigerants are two kinds of liquids that can be utilized for refrigeration. Essential refrigerants are liquids that are used straightforwardly as working liquids in refrigeration frameworks like fume pressure and fume pressure refrigeration. These liquids produce refrigeration in pressure or assimilation frameworks by going through a stage change in the evaporator. Auxiliary refrigerants are liquids that are utilized to convey nuclear power starting with one region then onto the next, as their name suggests. Brackish water or liquid catalyst are different names for auxiliary refrigerants. Assuming the functioning temperatures are above 0C, an unadulterated air blower may be used as a reinforcement. For instance, in huge cooling frameworks, refrigerant is utilized. The optional refrigerants are not exposed to similar tests as the essential refrigerants. As they communicate energy starting with one spot then onto the next, their stages shift starting with

one area then onto the next. The edge of freezing over of an auxiliary refrigerant is a significant element. The edge of freezing over of salt water is ordinarily lower than the edges of freezing over of its constituents. Temperature at which salt water arrives at its edge of freezing over is controlled by its fixation. The eutectic point is the focus at which the most minimal temperature can be gotten without hardening. The air blower arrangement with ethylene glycol, propylene glycol, or calcium chloride is a commonplace optional refrigerant. Brackish water arrangements are the conventional name for these arrangements. The functioning liquid in a fume assimilation framework is a refrigerant and spongy blend.

5.9.2 Refrigeration Selection Criteria

For a particular application selection of refrigeration is based on

- Thermodynamic and thermophysical properties
- Economics

Thermodynamics and Thermo Physical Properties

The necessities are

- Suction Pressure: When a specific measure of water has dissipated the immersion pressing factor ought to be equivalent to the temperature over the climate to forestall air contamination or dampness interruption into the framework, just as an instance of identification of a release a higher attractions pressure is gainful since it brings about a more modest blower dislodging
- Discharge Pressure: At a given condenser temperature the release pressing factor ought to be just about as little as conceivable to permit light weight development of blower and condenser.
- Pressure Ratio: It may be just about as little as workable for the high volumetric effectiveness as low force utilization.
- Latent and Heat of Vaporization: As extensive as conceivable so the necessary mass stream rate per unit cooling limit will be little

Environmental and Safety Properties

Environmental and safety factors are just as important as thermodynamic and thermos physical properties. In reality, at this time, the refrigerant's environmental friendliness is a crucial component in determining the usefulness of a certain refrigerant.

CHAPTER 6

LPG REFRIGERATION

This chapter includes

6.1 Physical Properties & Characteristics of LPG

6.2 Drawbacks of LPG

6.3 Components of LPG System

Refrigeration

HFC 134a is used as a refrigerant in more than 80% of residential refrigerators in Europe because of its outstanding thermodynamic and thermophysical qualities. On the other side HFC 134a has a high GWP. There is a need to weigh the pros and downsides of various refrigerant and alternative refrigerators that are currently in use and those that will be sold in the future. CFCs are primarily degraded because the chlorine produced in the high stratosphere catalysis the ultraviolet radiations in the stratosphere. Ultraviolet radiation penetrates to lower altitudes due to ozone degradation into oxygen. Car air conditioners have an impact on ozone layer that cannot be overlooked. HFCs (hydrofluorocarbons) are a type of fluorocarbon. HFCs like R-134a have high radiation qualities, hence they can't be used as a replacement agent of global warming. CFC 12 replacements have been reported using HFC 134a and the HC combination. They do, however, have drawbacks in terms of energy efficiency, flammability, and serviceability. Because mineral oil and HFC 134a are incompatible, It is suggested that you use polyol ester oil. It is highly hygroscopic in nature..

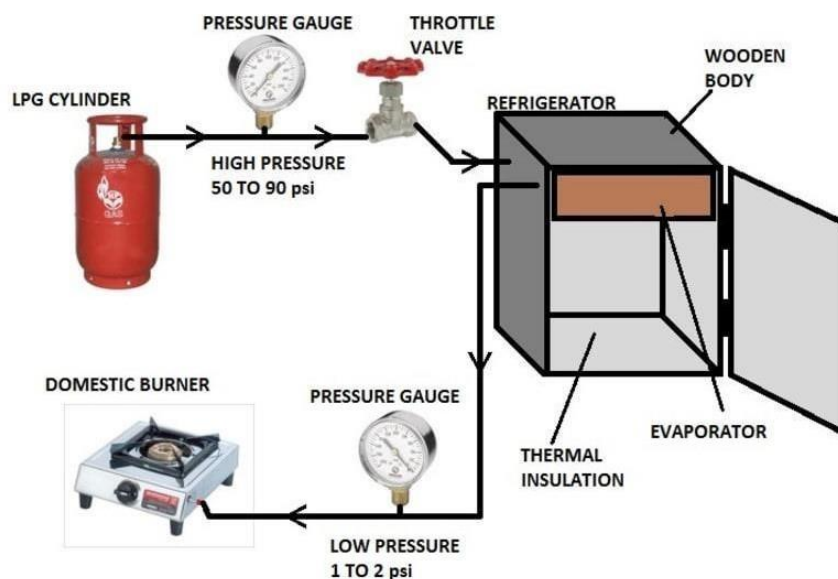


Figure 6.1 LPG Refrigeration

Liquefied Petroleum Gas

LPG (Liquefied Petroleum Gas) is a liquefied hydrocarbon gas found in melting structures. LPG is a dreadful, low-carbon, and incredibly efficient fuel. Propane (C_3H_8) and butane (C_4H_{10}) are the two basic structures (C_2H_8).

6.1 Physical Properties and Characteristic of LPG

1) Density

LPG is a gas that is 1.4 to 2.1 times heavier than air at atmospheric pressure and temperature. Under modest pressures, it easily liquefies. The density of the liquid is half that of water, ranging from 0.437 to 0.579 at 15 degrees Celsius.

Because LPG vapour is heavier than air, it tends to settle at ground level or in low-lying areas, accumulating in depressions.

2) Vapour Pressure

The pressuring factor inside an LPG accumulating vessel/chamber will be the same as the fume pressure in the capacity vessel, which is connected to the temperature of the LPG. Temperature, like the quantity of hydrocarbons in the mixture, influences fume pressure. Any additional growth of the fluid in the chamber will increase the chamber pressure by approximately 14 to 15 kg/sq.cm for each degree centigrade. This investigates the potentially dangerous situation that could arise from chamber overloading FLAMMABILITY (1.2.3). LPG in air has an unstable range of 1.8 percent to 9.5 percent volume of gas. This is far less than other commonly used vaporous energises. This demonstrates the possible risk of LPG fume gathering in lowlying regions on account of a break or spill. LPG's auto-start temperature is around 410-580 degrees Celsius, subsequently it will not light all alone at room temperature. During the siphoning/fillingin methodology, captured air in the fume can be hazardous in an unpurged vessel/chamber. Subsequently, utilizing pneumatic stress to off load LPG cargoes or big haulers isn't suggested.

3) Combustion

LPG combustion increases the volume of products in addition to creating heat. LPG requires up to 50 times the volume of air it contains to completely burn. Proper ventilation is necessary when LPG is used in enclosed spaces; otherwise, asphyxiation from oxygen depletion and the production of carbon dioxide may occur.

4) Odour

Because LPG has such a faint odour, it is important to add an odorant to help identify any escaping gas. Ethyl mercaptan is a frequent odour agent used for this purpose. The amount

to be added, according to IS: 4576, should be sufficient to allow detection in the environment at 1/5 of the lower limit of flammability, or fragrance level 2.

5) Colour

LPG is colourless in both its liquid and vapour phases. During a leak, liquid vaporisation cools the atmosphere and condenses the water vapour in it, generating a whitish fog that can be seen as an LPG escape.

6) Toxicity

LPG, while slightly toxic, is not harmful in the vapour phase because it displaces oxygen. It can cause suffocation in high amounts. As a result, the vapour has modest anaesthetic characteristics.

6.2 Drawbacks of LPG

- When in a gaseous state, LPG is twice as heavy as air and will generally descend to the lowest possible level, accumulating in basements, pits, sewers, and other locations.
- LPG in a fluid structure can give extreme viral consumes to the skin due to its rapid vaporisation.
- Vaporization can cool hardware to where it can cause cold consumes.
- In focuses going from 2% to 10%, LPG makes a combustible combination with air.
- As an outcome, whenever put away or dealt with improperly, it very well may be a fire and dangerous risk.
- The fire can head out back to the wellspring of the break if fume/air blends arising out of holes are lighted some separation from the place of getaway.
- Despite the way that a vessel that has recently held LPG is apparently unfilled, it might in any case contain LPG fume and be hazardous.
- It may be risky. Thus, treat all LPG vessels like they are totally full.

6.3 Components of LPG Refrigeration System

1) LPG Cylinder

Butane and isobutene are combined as one to make LPG. It is usually kept at 12.7 bar for household purposes. With the help of a decent controller, LPG is injected into a thin cylinder. LPG is used in a variety of ways, including homegrown, mechanical, plant, rural, cooking, warming, and drying. Among other things, LPG can be used as a car fuel or an airborne charge. Through the use of pressing factor lamps, LPG can also be used to provide lighting.



Figure 6.2 LPG Cylinder

2) Capillary Tube

In family refrigeration, the slim cylinder is the most mainstream choking gadget. A copper tube with a minuscule interior breadth is utilized as a fine cylinder. It has an extremely protracted length and is wound into different goes to occupy less room. The fine cylinder utilized for refrigeration applications has an inside distance across going from 0.5 to 2.28 mm (0.020 to 0.09 inch). The fine cylinder is portrayed in the delineation. The distance across of the hairlike and the length of the fine decide the decrease in refrigerant pressing factor by means of the fine. The lower the pressing factor of the refrigerant as it courses through the hairlike cylinder, the more modest the distance across and the more drawn out the length.



Figure 6.3 Capillary Tube

3) Evaporator

Another important component of refrigeration systems is evaporators. The cooling impact in the refrigeration framework is created by the evaporators. In refrigeration frameworks, the real cooling impact happens in the evaporators. Numerous individuals accept the evaporator to be the main part of the refrigeration framework, while different segments are considered less significant. Evaporators are heat exchanger surfaces that transfer heat from a cooled material to the refrigerant while also displacing heat from the substance.



Figure 6.4 Evaporator

4) Pressure Gauge

For the measurement of pressure and vacuums, a variety of techniques have been devised. Pressure gauges and vacuum gauges are instruments that are used to measure pressure



Figure 6.5 Pressure Gauge

5) High Pressure Pipes

The high-pressure pipe series is suitable for the majority of applications requiring high-pressure gas transfer. They're made out of a steel pipe with steel balls on both ends. These balls are squeezed against the seating of the interfacing opening by two turning association areolas, forestalling gas spillage. All lines are pressure tried to a functioning pressing factor of 100 M Pa (14,500 psi).



Figure 6.6 High Pressure Pipe

6) High Pressure Regulator

This type of controller is used to remove gas with a high pushing factor from the chambers. These are mostly used in the capacity of Bhatti ovens.



Figure 6.7 Regulator

CHAPTER 7

CONSTRUCTION AND WORKING

This chapter includes

7.1 Construction of LPG Refrigeration

7.2 Working of LPG Refrigeration

7.3 Advantages & Disadvantages of LPG Refrigeration System

7.4 Application of LPG Refrigeration System

7.5 Result Verification

7.6 Conclusion

7.1 Construction

The LPG cooler is portrayed in the outline. The Thermo-coal sheet is made in one thermo-coal sheet within the refrigerator. The evaporator is additionally totally canvassed in aluminum tape. The LPG refrigeration framework is portrayed in the chart beneath. This is the place where the gas bundle. For the LPG cooler, the thermo-coal sheet size is 15mm. The evaporator's measurements are 355*254*152 mm³. Since cold air can't move from inside to outside, we safeguarded the chamber is associated. High-pressure pipes are connected to a high-pressure controller. A pressing factor check is related with the lines on the opposite side of the extreme pressing factor. A copper tube is connected to the other end, which is connected to the thin cylinder. An evaporator is installed in the fine cylinder. Another high pressing factor pipe connects the evaporator loop end to the oven. One pressing factor measure is placed between the fine cylinder and the chamber, while the other is placed towards the evaporator's end.



Figure 7.1 Construction of LPG Refrigeration

7.2 Working

The core concept of the LPG fridge is to use LPG to preserve heat. In the graph below, the functioning instrument of LPG refrigeration is depicted..

- In an LPG chamber, LPG is stored under high strain. When the valve is opened, a stream of high-pressure LPG flows through the high-pressure pipe.
- At the slim cylinder, high-pressure LPG is changed over to low-pressure LPG while the enthalpy stays steady.
- As a result, the chamber cools down after passing through the evaporator, and low-pressure LPG passes via the line to the burner, creating a chilling effect in the fridge.
- We can likewise utilize LPG's low pressing factor in consuming techniques.

The Simple Vapour Compression Refrigeration system is used in the LPG Refrigerator. The following is how the VCR system works:

Process 2-3: When the blower is turned on, it pulls low-pressure fume from the evaporator in state 2 and packs it isentropically to a sufficiently high pressing factor in state 3. Because the fume is subjected to pressure, its temperature rises, and it is subsequently converted to a low pressing factor adiabatically, such that the enthalpy remains constant.

This low-pressure LPG passes through an evaporator after narrowing in the cylinder. The chilling chamber cools as a result of the evaporation. Tor LPG is converted into a low-pressing-factor, low-temperature structure that holds the heat from the cooling chamber.

Process 3-4: The hot fume from the blower is sent into the condenser, where it is ingested by the condenser cooling medium, which is usually water or surrounding air. The heated fume is converted to fluid in state 4, and the fluid is collected in a fluid collector.

Process 4-1: High fluid from the fluid collector is directed to a refrigerant control valve, which control fluid stream into the evaporator. While confining the flow, this control valve likewise decreases the pressing factor o the liquid, causing the fluid to change over into a low dryness portion which is addressed by state 1. The temprature of the refrigerant drops in propotion to its pressing factor during this activity.

Process 1-2: At long last, the low-pressure, low-temperature refrigerant courses through the evaporator loop, engrossing idle warmth from the virus chamber or brackish water arrangement at steady pressing factor and changing over to fume at state 2. It's given to the blower again. The cycle is presently finished.

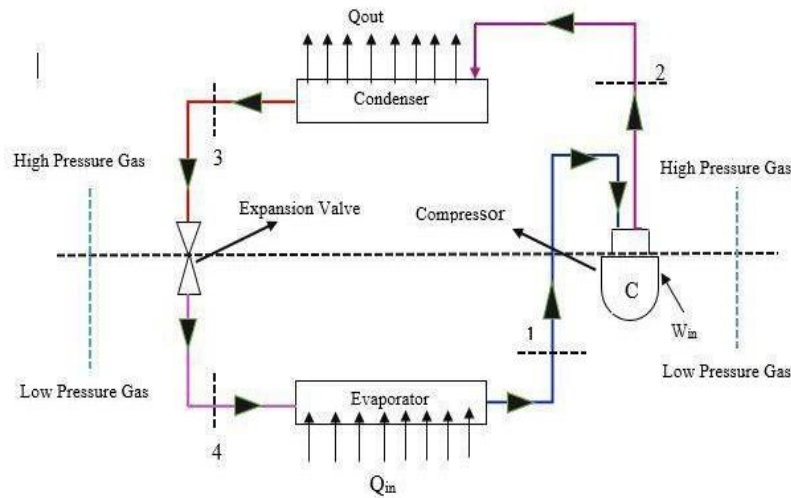


Figure 7.2 Working Principle of LPG Refrigeration

The idea driving LPG refrigeration is to ingest heat from the climate by dissipating a LPG. The pressing factor of the LPG in the chamber is around 80 psi. We are utilizing slender to drop the pressing factor of LPG to 15 psi, permitting the climate to be cooled by engrossing warmth isentropically.

The pressing factor of LPG in the chamber is high, and when the fuel tank's controller is opened, highpressure LPG goes through the gas line. The high-pressure LPG is then infused into the slender cylinder through the high-pressure pipe. In view of the development of the LPG gas in the slender cylinder, the high-pressure LPG is changed to low pressing factor and thus low temperature.

As a result, we can expect a cooling effect in the cooler. From there, the low-pressure LPG from the evaporator is sent to the burner via a high-pressure pipe, and we can use this low-pressure LPG for consumption. Rather than utilizing a blower, we utilized a recompressed LPG chamber in this venture. We can produce a refrigerating impact from this framework along these lines.

7.3 Advantages and Disadvantages

Advantages

- Using LPG as a refrigerant improves total efficiency by ten to twenty percent.
- LPG has no ozone depletion potential.
- Aside from being environmentally beneficial, using LPG has a number of economical benefits.
- Because of the higher density of LPG, the system's weight is reduced by 60%.

- This refrigerator functions even when the power is turned off.
- In operation, the pieces are virtually silent.
- The compressor and condenser are no longer needed, resulting in a zero-cost operation.

Disadvantages

- The level of efficiency is low.
- The explosion is caused by LPG leakage.
- The system is tough to repair and maintain.
- The system is quite large.

7.4 Applications

- It has the potential to be very useful in restaurants that demand constant cooling and heating.
- It can be handy in isolated areas without access to electricity.
- It can be utilized in refineries with heavy LPG usage.
- The system is suitable for industrial central cooling as well as home refrigeration and air conditioning.
- • It can be used in automobiles that run on LPG or other gaseous fuels for air conditioning.
- It can be handy in isolated areas without access to electricity.
- It is simple to cool and store essentials in isolated places and emergency vehicles, such as important bio-chemicals, injections, and other medical supplies in an ambulance.
- It can be utilized for free air conditioning in locations with their own gas turbine power plants, such as airports, retail malls, and so on.

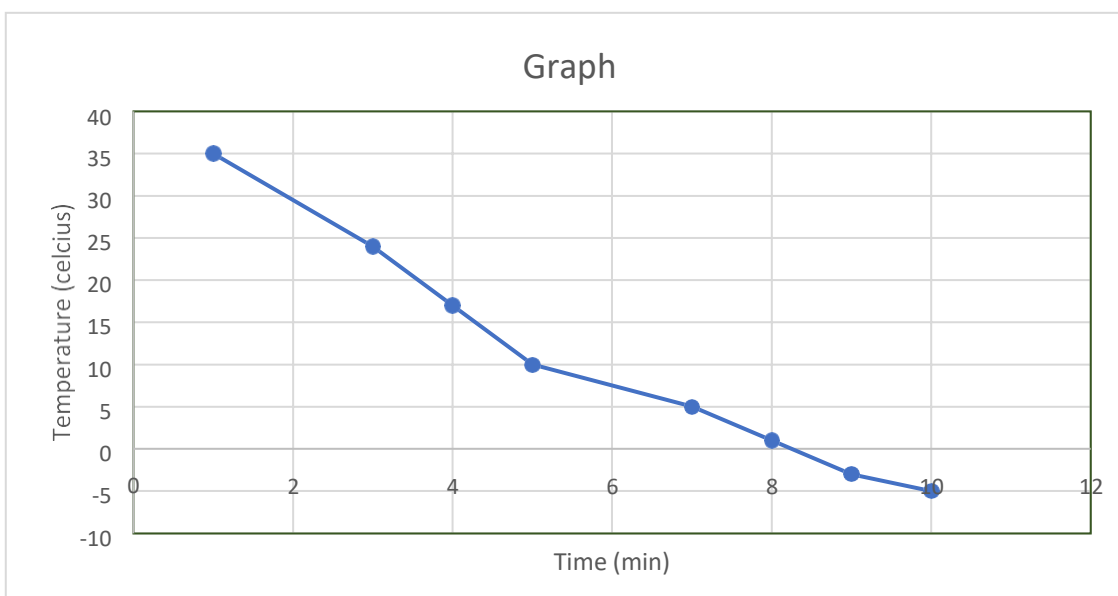
7.5 Result Verification

During experiment we have analyzed the sudden drop of temperature of evaporator from 35°C to -5°C in just 5 to 10 minutes. Table is given below of certain drop of temperature in different time duration.

Table 7.1 Result verification

Sr. No.	Time (s)	Temperature (°C)
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1	1	35
2	3	24
3	4	17
4	5	10
5	7	5
6	8	1
7	9	-3
8	10	-5
9	12	-5
10	15	-5



7.5 Conclusion

The goal of the LPG cooler was to use LPG as a refrigerant and to achieve the refrigerating effect by utilising the energy of the chamber's great pressing factor. In a Domestic 14.5 kilogramme chamber with a high-pressure controller, we have LPG at a pressing factor of 12.41 bar, which has been reduced to 1.41 bar with the help of a slim cylinder. Notwithstanding, on the off chance that we utilize a low-pressure controller, as is normal in private LPG gas ovens, the LPG pressure after the extension gadget and before the burner would be unique. Subsequently, we processed the refrigerating impact utilizing changes in LPG qualities (pressing factor, temperature, and enthalpy) prior and then afterward the evaporator utilizing high pressing factor and high temperature. The amount of refrigerating effect is determined utilizing a pressing factor controller. The COP will increment with this energy input. LPG cooler has a rating of 5.08, which is higher than the homegrown fridge. On the off chance that the energy contribution for 1 kg of LPG producing was gotten from the energy review, the outcome may be unique. This framework is more affordable to set up

and keep up with. It doesn't require the utilization of outside fuel sources. There are no moving parts in the framework, accordingly it tends to be run. Thus, the expense of support is likewise very insignificant. This framework is compelling. A large portion of the ideal for inns, ventures, treatment facilities, and substance plants with hefty LPG utilization.

We can conclude that

In contrast to the CFCs now in use, propane is an appealing and environmentally friendly alternative. In fine inward measurement, the mass stream rate increases with expansion, but it decreases with lengthy expansion. It was discovered that the COP of the framework increases in proportion to the change in the slender cylinder calculation. The cooling limits for LPG were roughly three to four times higher than those for R-12. The COP of an LPG refrigerator was 8.9% greater than that of R134a. LPG appears to be a viable long-term replacement for R134a in the existing cooler. The concentrations of carbon monoxide (COP) were abnormally high. There have been no issues with the blower's activity. The utilization of LPG as a refrigerant substitution can assist with addressing the (ODP) issue and decrease a dangerous atmospheric deviation potential.

REFERENCES

1. Arora, "Refrigeration and Air conditioning", capillary tube design (2010) 311-317.
2. Dr. Amir S. Dawood and Salim Ibrahim Hasan, "Numerical study of Refrigerant flow in capillary tube using Refrigerant (R134) Journal (2011) 119
3. S. Zainazakria and Zuliaikhashahrum, "The possibility of using LPG in Domestic Refrigeration System" (2011) 347-354.
4. Wong, T. N. Ooi, K.T. "Adiabatic capillary tube expansion devices: comparison of the homogeneous flow and the seperated flow models" Applied Thermal Eng.16 (7) (1996) 625634.
5. Bilal, A. A. and Salem, A. A. (2002). Assessment of LPG as possible Alternative to R12 in Domestic Refrigerators. Ener. Conv and Man, 44: 381 388. Fatouh, M. and Kafafy, M.E (2005). Experimental Evaluation of a Domestic Refrigerator Working with LPG.26: 15931603.